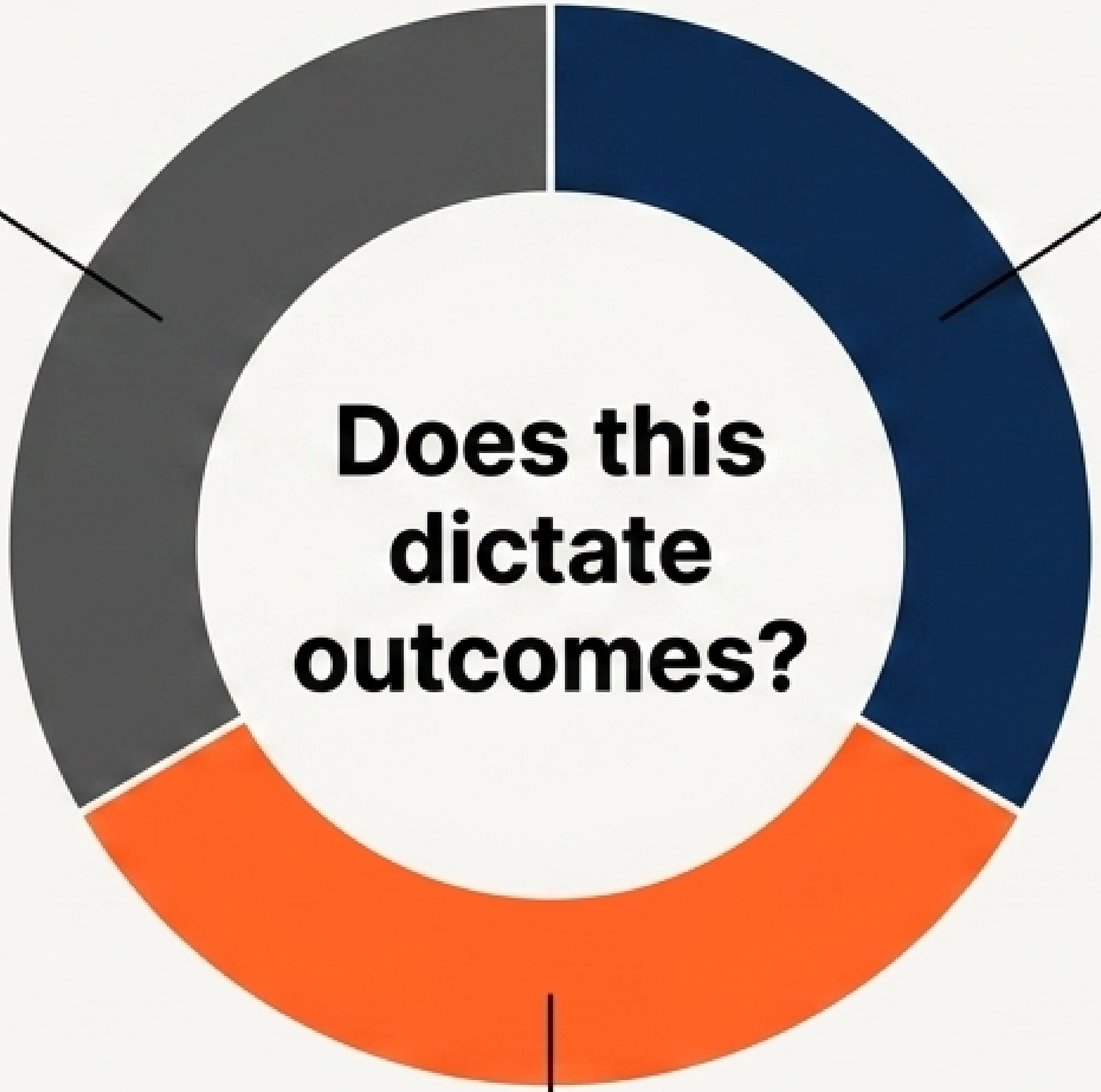
The Fatigue Factor

STAT 248 Time-Series Analysis | Final Project
By Chenfei Peng



Travel Load

Circadian rhythm disruption and late-night flights across time zones.

Physical Exhaustion

The physiological toll of playing 48 minutes of professional basketball.

Mental Toll

Focus degradation and game plan execution errors.

- Do back-to-back games universally penalize teams?
- Can ARIMAX capture momentum better than standard regressions?
- Can we quantify the exact point penalty out-of-sample?

Mapping the Data Landscape

Time Variables

Capturing schedule density, specifically tracking the boolean flag is_back_to_back.

Rest Variables

Tracking is_short_rest conditions and days elapsed since the last game.

Travel Variables

Quantifying geographic displacement between consecutive road games.

Base API Metrics

Importing point differentials and opponent strength from nba_api.

NotebookLM

Escalating Statistical Rigor

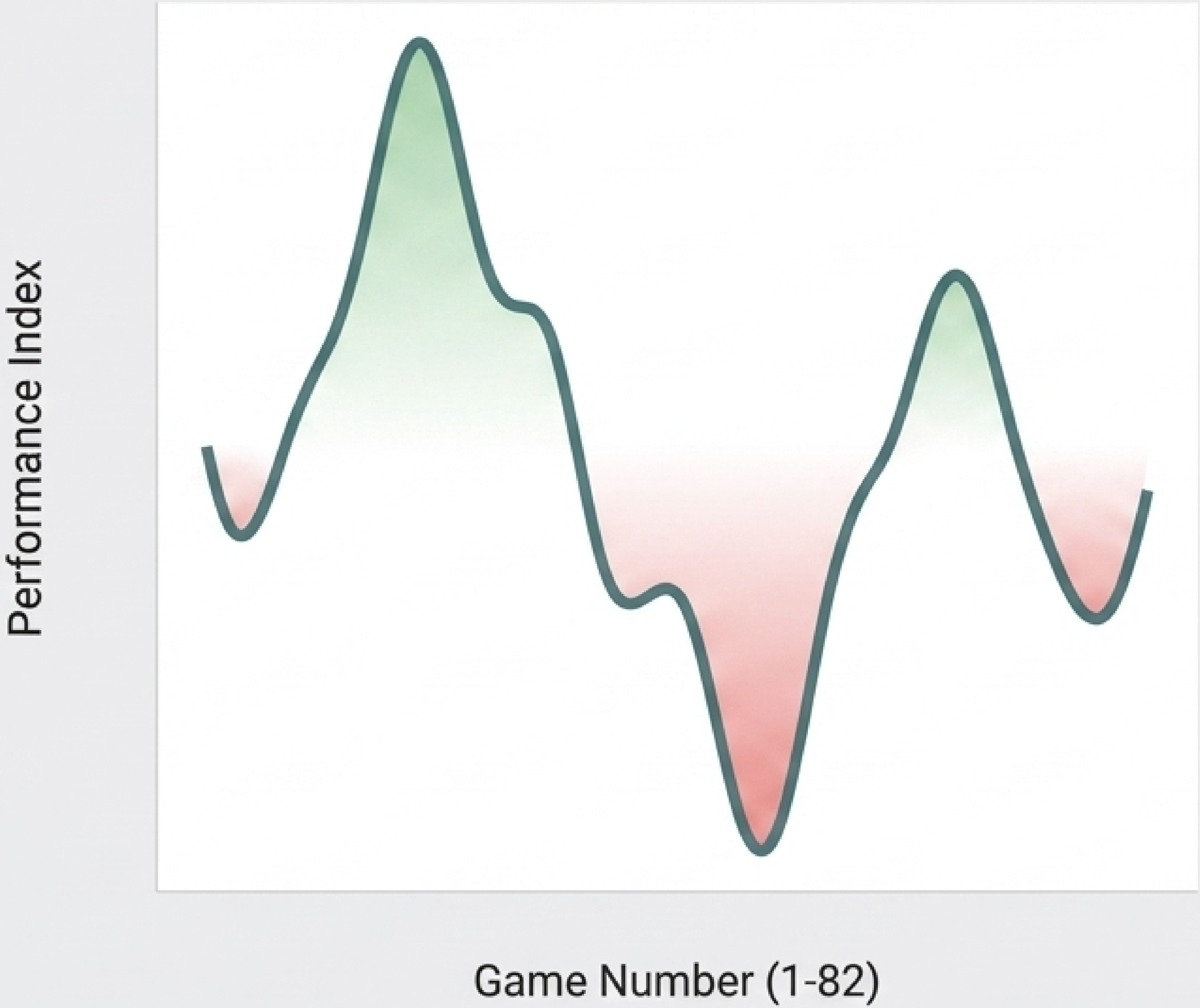
Progressing from basic global correlation to rigorous out-of-sample predictive validation.

Establishing the Global Baseline Correlation

Variable	Coefficient	Std Error	t-Statistic	p-Value
Intercept	-1.897	0.657	-2.88	0.004
team_strength	0.892	0.045	19.82	<0.001
is_back_to_back	-2.802	0.582	-4.81	< 0.001
opp_is_back_to_back	+2.740	0.472	5.80	< 0.001
travel_dist_1k	-0.180	0.090	-2.00	0.045

Initial Pooled OLS confirms a universal, statistically significant point penalty for playing on zero days rest.

Isolating True Signal by Modeling Team Momentum



$$\text{SARIMAX}(1,0,0)$$

$$y_t = \alpha + \phi_1 y_{t-1} + \beta X_t + \varepsilon_t$$

OLS assumes games are independent. In reality, basketball relies on momentum (streaks and slumps). ARIMAX isolates true fatigue from a bad slump by capturing the ‘stickiness” of a team’s performance.

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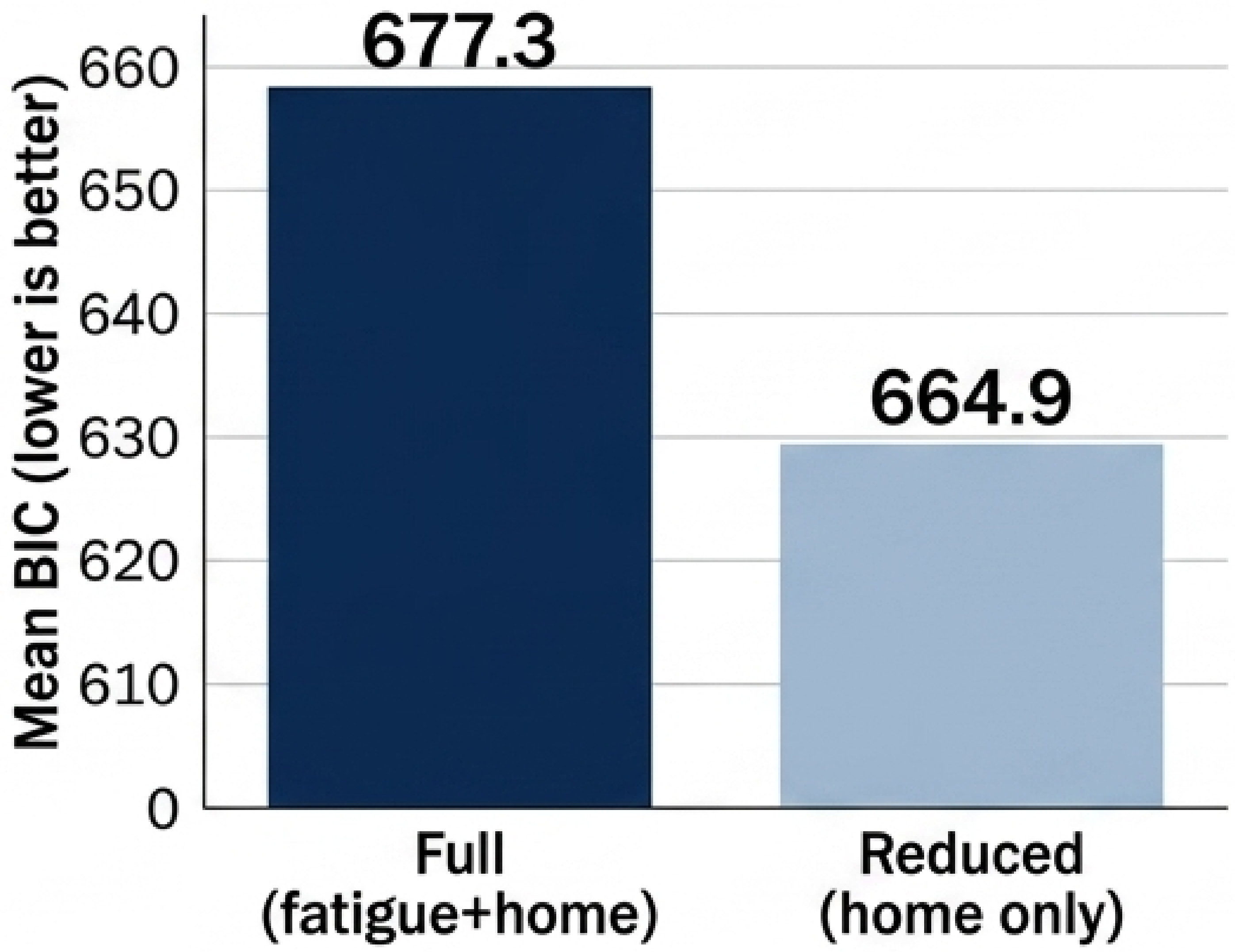
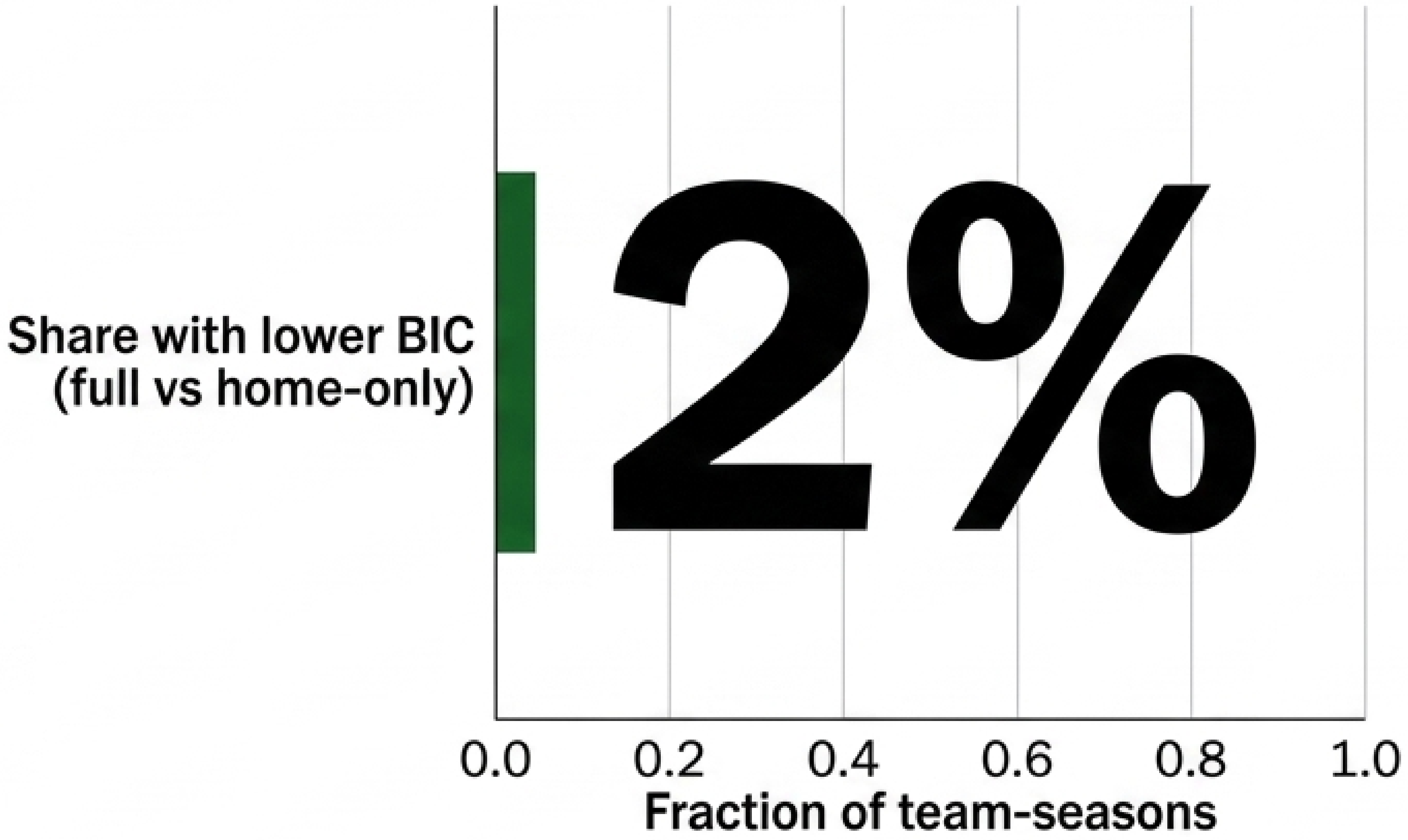
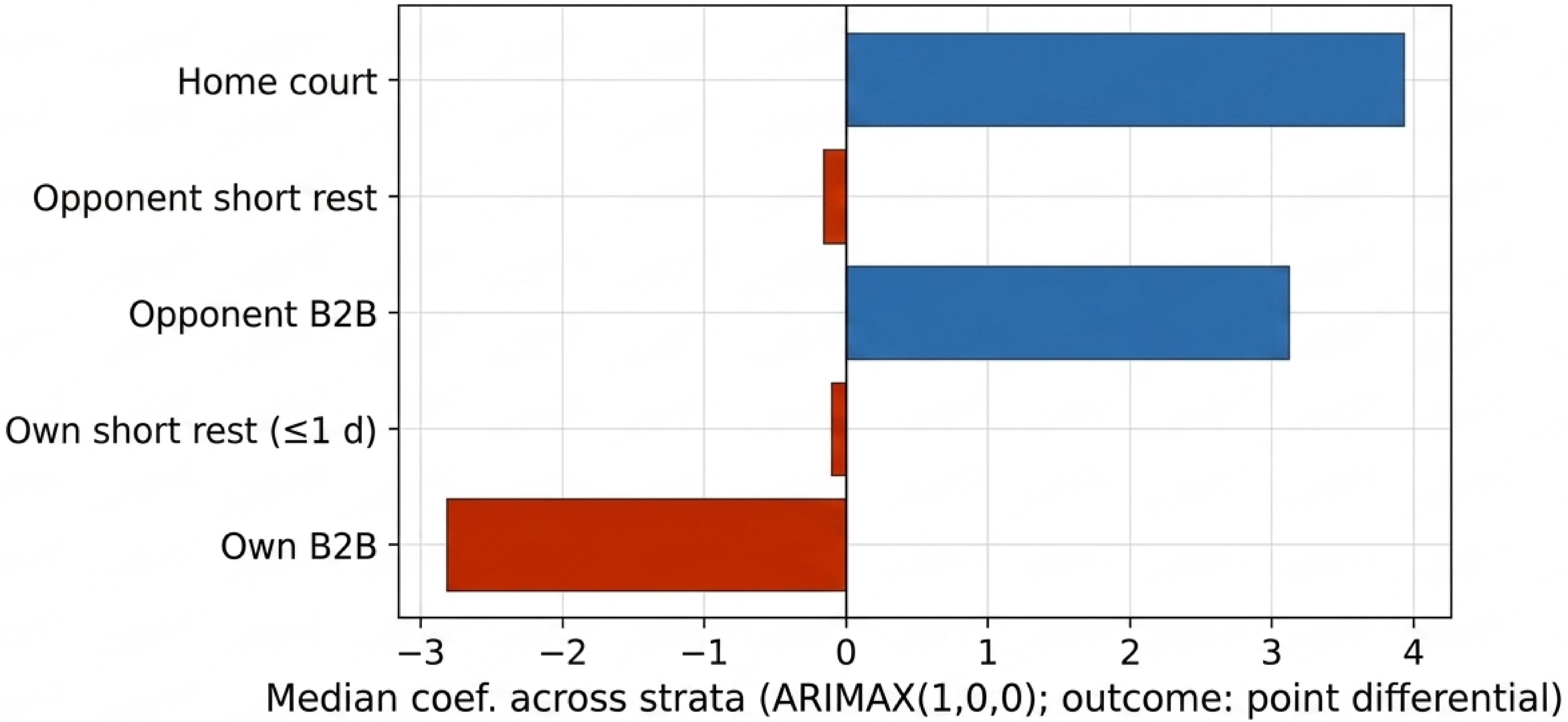
The Strict Mathematical Penalty of ARIMAX

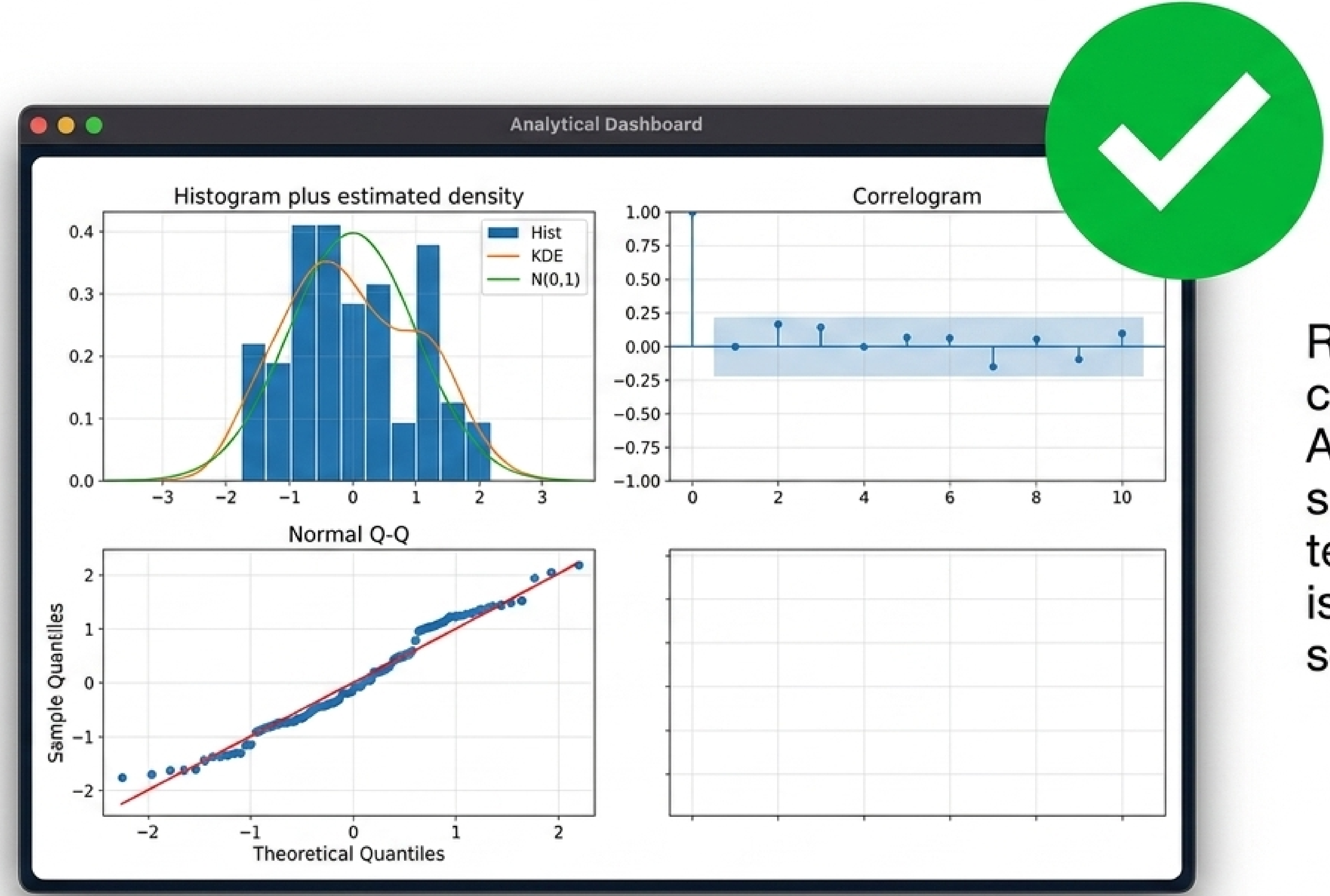




Due to the strict penalty for adding parameters in BIC, the full model (home + fatigue) only definitively out-performed the home-only baseline in 2% of team-seasons.



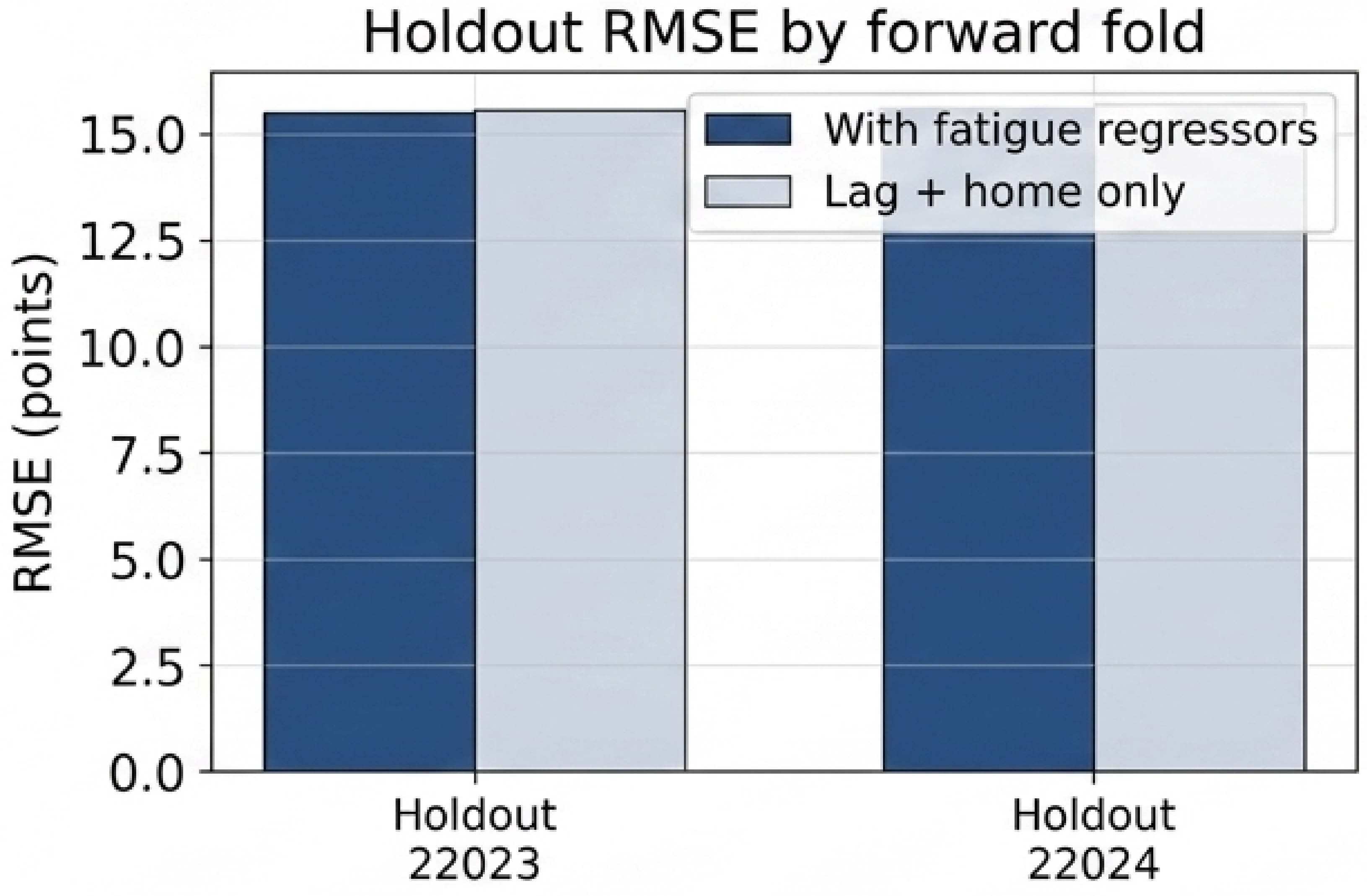
ON AVERAGE, A B2B COSTS A TEAM 2.82 POINTS ON THE MARGIN.



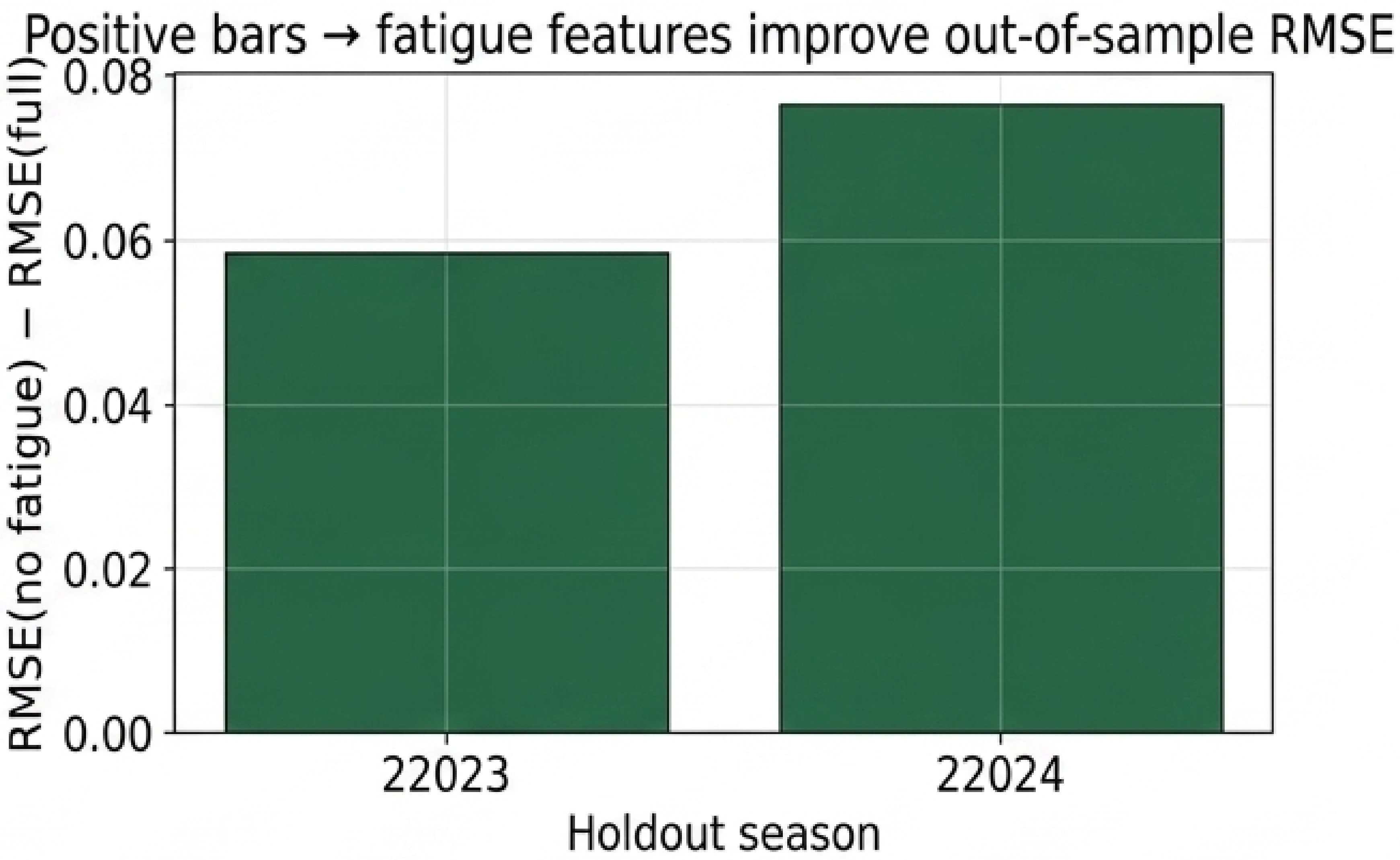
Residual ACF/PACF plots confirm White Noise. The AR(1) component successfully captured team momentum, isolating the true schedule penalty.

Validating Predictability with Forward Seasonal CV

Holdout RMSE by forward fold

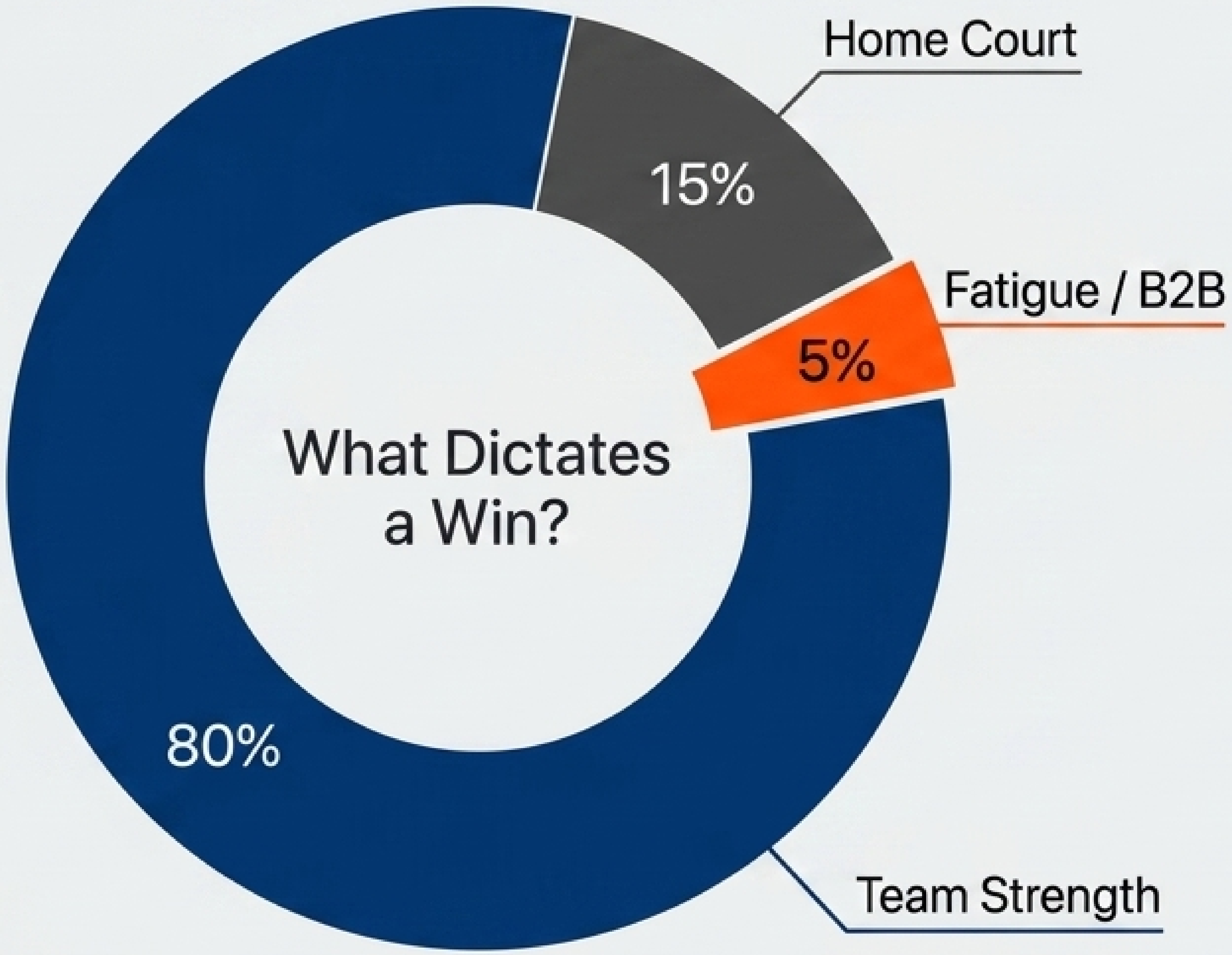


Predictive Gain Over Baseline



Fatigue predictors consistently reduce RMSE in forward-looking seasonal tests.

Putting Fatigue in Perspective



Fatigue is undeniably significant and costs ~2.8 points, but it is a marginal factor (~5% impact).

Baseline team strength (80%+) remains the primary driver of victory. Fatigue only decides the closest of contests.

Uncaptured Variables in the 82-Game Grind

Load Management

The API cannot distinguish true exhaustion from the strategic resting of star players.

True Distance

Metrics track point A to point B, completely missing the physical toll of actual air miles flown or weather-related tarmac delays.

The Injury Noise

Roster depletion often masquerades as schedule-induced fatigue in the macro data.

THANK YOU.

Questions?

https://github.com/ChenfeiP/248-final-project.git